

ANSWER KEY — Integration Lesson 28 Test: Reading

Accumulation Graphs

For the grader · full working shown · justification questions score ONLY if the sentence names $g' = f$ and its sign change ("graph goes down" = no credit) · exact answers required where π appears

GRAPH 1 (questions 1–7). f on $[0, 8]$: rises $(0, 0) \rightarrow (2, 4)$; flat at 4 from $t = 2$ to 5; straight line down from $(5, 4)$ to $(8, -2)$, crossing zero at $t = 7$. $g(x) = \int_0^x f(t) dt$. Useful slopes: first segment slope 2; last segment slope -2 , with $f(t) = 4 - 2(t - 5)$ on $[5, 8]$.

1. Find $g(2)$.

Work: region from 0 to 2 is a triangle: base 2, height 4 $\rightarrow \frac{1}{2} \cdot 2 \cdot 4 = 4$, above the axis so positive.
Answer: $g(2) = 4$.

2. Find $g(5)$.

Work: $g(5) = g(2) + \text{rectangle from 2 to 5: base 3, height 4} \rightarrow 12$. So $4 + 12 = 16$. **Answer:** $g(5) = 16$.

3. Find $g(7)$ and $g(8)$.

Work: from 5 to 7, f falls from 4 to 0: triangle, base 2, height 4 $\rightarrow +4$, so $g(7) = 16 + 4 = 20$. From 7 to 8, f falls from 0 to -2 : triangle, base 1, height 2 $\rightarrow$ area 1, BELOW the axis $\rightarrow -1$, so $g(8) = 20 - 1 = 19$. **Answer:** $g(7) = 20$ and $g(8) = 19$. (Common error: forgetting the minus on the last piece $\rightarrow 21$.)

4. Find $g'(1)$ and $g'(6)$.

Work: FTC 2: $g'(x) = f(x)$ — read heights off the graph. $f(1) = 2$ (halfway up the segment from $(0,0)$ to $(2,4)$, slope 2). $f(6) = 4 - 2(6 - 5) = 2$ (one unit along the falling segment). **Answer:** $g'(1) = 2$ and $g'(6) = 2$.

5. Where is g increasing / decreasing? Justify.

Work: $f > 0$ on $(0, 7)$ (above the axis until it crosses zero at $t = 7$); $f < 0$ on $(7, 8)$. Model sentence: "g is increasing on $(0, 7)$ because $g' = f$ is positive there, and decreasing on $(7, 8)$ because $g' = f$ is negative there." **Answer:** increasing on $(0, 7)$; decreasing on $(7, 8)$. **Grader note:** the sentence must tie the behavior of g to the SIGN of $g' = f$.

6. Relative maximum of g ? Justify, and give the value.

Work: " g has a relative maximum at $x = 7$ because $g' = f$ changes from positive to negative there." Value: $g(7) = 20$ (from question 3). **Answer: relative max at $x = 7$, where $g(7) = 20$.** **Grader note:** "because f is highest at $x = 7$ " is WRONG reasoning (f is highest on $(2,5)$) — no credit without the sign-change sentence.

7. Concavity of g ? Justify.

Work: $g'' = f' = \text{slope of the pictured graph}$. On $(0, 2)$, f is increasing $\rightarrow g$ concave up. On $(2, 5)$, f is constant $\rightarrow g$ is a straight line (slope 4), no concavity. On $(5, 8)$, f is decreasing $\rightarrow g$ concave down. Sentence: " g is concave up on $(0, 2)$ because $g' = f$ is increasing there, and concave down on $(5, 8)$ because $g' = f$ is decreasing there." **Answer: concave up on $(0, 2)$; concave down on $(5, 8)$; straight line on $(2, 5)$.**

GRAPH 2 (questions 8–10). f on $[0, 10]$: half-circle of radius 2 above the axis from $t = 0$ to 4; line down from $(4, 0)$ to $(6, -2)$; flat at -2 from $t = 6$ to 10. $g(x) = \int_0^x f(t) dt$.

8. Find $g(4)$, exactly.

Work: half-circle area $= \frac{1}{2}\pi r^2$ with $r = 2 \rightarrow \frac{1}{2} \cdot \pi \cdot 4 = 2\pi$, above the axis so positive. **Answer: $g(4) = 2\pi$.** (Exact form required; 6.28 alone loses the point.)

9. Find $g(6)$ and $g(10)$, exactly.

Work: from 4 to 6: triangle, base 2, height 2 $\rightarrow$ area 2, below the axis $\rightarrow g(6) = 2\pi - 2$. From 6 to 10: rectangle, base 4, height 2, below the axis $\rightarrow -8 \rightarrow g(10) = 2\pi - 2 - 8 = 2\pi - 10$. **Answer: $g(6) = 2\pi - 2$ and $g(10) = 2\pi - 10$.**

10. (a) Exact $x > 0$ with $g(x) = 0$. (b) $h(x) = 1 + \int_0^x f(t) dt$: find $h(4)$.

Work (a): $g(6) = 2\pi - 2 \approx 4.28 > 0$ and $g(10) = 2\pi - 10 \approx -3.72 < 0$, so the zero lies on the flat piece $(6, 10)$. There g falls at a steady 2 per unit: $g(x) = g(6) - 2(x - 6) = 2\pi - 2 - 2(x - 6)$. Set to 0: $2(x - 6) = 2\pi - 2 \rightarrow x - 6 = \pi - 1 \rightarrow x = 5 + \pi \approx 8.14$ ✓ (inside $(6, 10)$). Check: $g(5 + \pi) = 2\pi - 2 - 2(\pi - 1) = 0$ ✓. **Work (b):** h is g shifted up 1, so $h(4) = 1 + g(4) = 1 + 2\pi$. **Answer: (a) $x = 5 + \pi$ (b) $h(4) = 1 + 2\pi$.** **Grader note:** for (a), give the point only for the exact form $5 + \pi$ (or equivalent); a decimal ≈ 8.14 alone earns half.